

HAT-P-36b

R-0936

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230320 · created 2026-07-29T12:14:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460023.7311 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.043
Transit depth (Rp/Rs)^2	1.37 +/- 1.01 [%]
Orbital Inclination (inc)	88.27 +/- 2.81
Airmass coefficient 1 (a1)	0.904 +/- 0.021
Airmass coefficient 2 (a2)	0.091 +/- 0.07
Scatter in the residuals of the lightcurve fit is	2.3 %
Best Comparison Star	#1 - [138, 149]
Optimal Method	PSF photometry
Transit Duration (day)	0.0647 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.117 ± 0.043 vs 0.126 (deviation 0.21σ)
Duration fit vs published	0.0647 d vs 0.0925 d (deviation 4.96σ)
Mid-time O-C	-12.6 min
Depth SNR	2.7
Residual scatter	2.3 %

Light curve

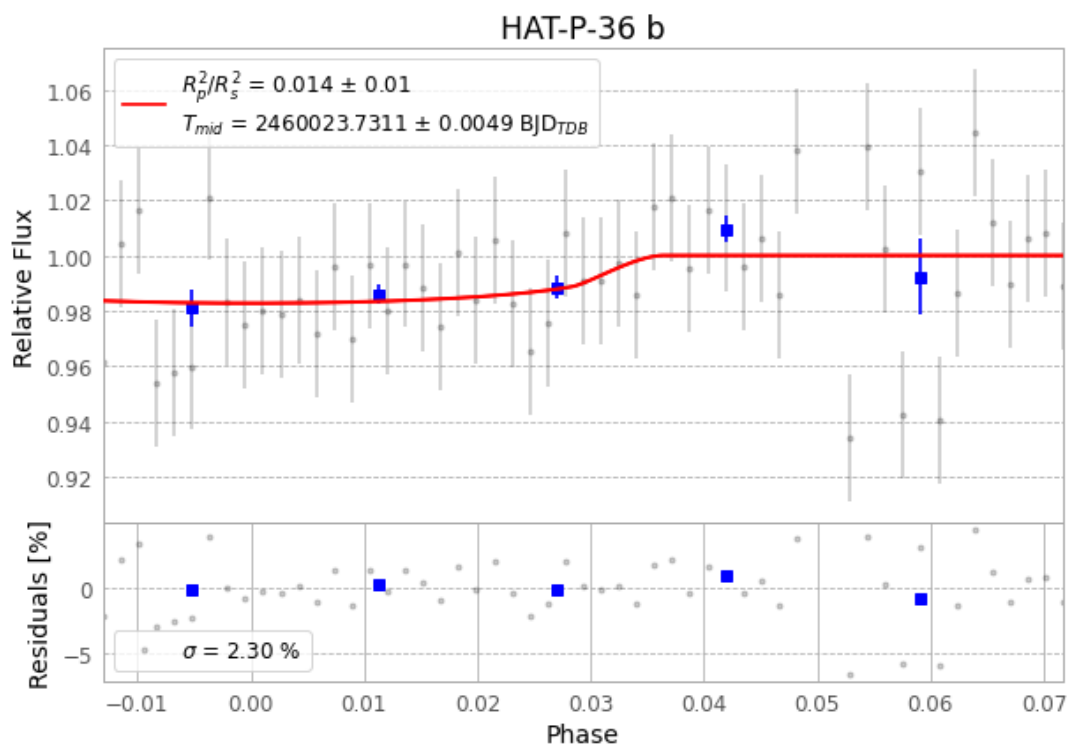


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-03-19.png (SHA-256 34d41e1b8eb250cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [536, 114], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1124c98150e03be3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

53 FITS frames (35 MB), HATP-36230320050016.FITS → HATP-36230320074216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230320.log (8cb867c103bb...), FinalLightCurve_HAT-P-36 b_2023-03-19.png (34d41e1b8eb2...), FinalLightCurve_HAT-P-36 b_2023-03-19.csv (621a37922e4a...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 12:14Z.